Physics-Informed Machine Learning for Predicting UV-Vis Spectra

Candidate Number: 1102030

[1]Physical and Theoretical Chemistry Laboratory, University of Oxford, Oxford, United Kingdom

Abstract

Fast and accurate prediction of UV-Vis spectra remains challenging because of two primary limitations: (i) the scarcity of publicly available spectral data and (ii) the substantial computational cost associated with *ab initio* spectrum calculations. Here, we present an efficient machine-learning workflow that first performs an inexpensive excited-state and oscillator-strength calculation. A multilayer perceptron (MLP) is then trained to correct the calculated values, which are combined with molecular descriptors derived from SMILES representations to provide additional structural information. Finally, a random forest model uses these features to predict the UV-Vis absorption spectra of organic molecules. Across multiple random forest configurations, the best-performing models achieved root mean square errors (RMSEs) of 0.1066 for Dataset I and 0.0715 for Dataset II. These results were obtained using a relatively small training set of 3,063 molecule-spectrum pairs, demonstrating the potential of combining inexpensive quantum-chemical calculations with molecular descriptors and machine-learning corrections for efficient UV-Vis spectrum prediction.

Keywords: UV-Vis spectroscopy, machine learning, random forest, molecular descriptors, excited-state calculations, spectral prediction, computational chemistry

1. INTRODUCTION

Determining a molecule's optical absorption and emission properties is essential in screening for suitability in photovoltaic cells, drug design and organic light-emitting diodes (OLEDs). In performing this computational screening, chemists can be more targeted and resourceful in what chemicals they choose to synthesize. However, the ab-initio determination of UV-Vis spectra has so far proven difficult [3], since a full calculation on either an absorption or emission spectrum runs into intractable scaling laws. In summary, the first step to deducing an absorption spectrum is to calculate a molecule's excitation energies and oscillator strengths together, creating a "stick" spectrum. Doing this with Full CI scales combinatorially with the number of basis sets, making it non-viable for medium-large organic molecules. Practical alternatives (e.g. CASSCF, CC2, CISD(T)) have polynomial scaling of $\sim \mathcal{O}\left(N^{5\text{–}8}\right)$, which are still expensive. The next step is to evaluate vibrational modes, whereby a PES has to be constructed by evaluating total molecular energy (with one of these methods above) over roughly N_{atoms} configurations [1], thereby scaling with $\sim \mathcal{O}\left(N_{\text{atoms}} N^{5\text{–}8}\right)$. Finally, when quantifying solvent effects, we need to add the basis functions of solvent atoms into the quantum calculation, thereby scaling with:

$$\mathcal{O}\left((N + N_{\text{solv}})^{5\text{–}8}\right).$$

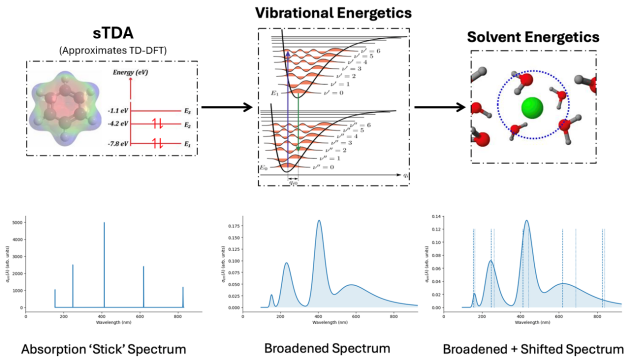


Figure 1. Flowchart to illustrate how a UV-Vis spectrum can be determined ab-initio, and how each method component contributes to a spectrum's shape.

As such, some attempts have been made to machine-learn UV-Vis spectra, with either simple SMILE structures as input [2] or full ab-initio spectra [3], SMILEs (Simplified Molecular-Input Line-Entry System) being a single text line descriptor of molecular structure. The model developed in this project aims to be a middle ground, essentially providing the model with some physics information (cheap excitation energies, oscillator strengths via sTDA) and let it machine-learn the spectra from there. As such, spectra can still be predicted in seconds-per-molecule but are nonetheless more accurate than predictions from plain molecular structures. This way, our model can be used to perform reasonably accurate high-throughput screening of candidate OLED molecules.

As is covered in the 3. METHODOLOGY AND DATA section in more detail, this model essentially runs a fast quantum calculation for the excited energies and oscillator strengths of the five energetically lowest excited transitions. Then, it learns to correct this 10-vector via a multi-layer perceptron (MLP) head, with reference to a database of 10,000 excited states at a qsGW-BSE level of theory. Then separately, in the same fashion as [2], we predict a molecule's absorption spectrum directly from its SMILES structure string, which we encode into a 1024-dim vector. Given this predicted spectrum, a random forest model learns to use this data in conjunction with the 10-dim quantum information vector, to apply a correction to the predicted spectrum, thereby yielding a final prediction.

2. PROJECT AIMS

The primary aim of this project is to create a model which can predict the absorption spectra of given organic molecules, within a timescale that makes it viable for high-throughput screening of candidate OLEDs, and do this to a comparable accuracy to the best available models in the literature [2] [3]. More specifically, we will use per-molecule runtime and root mean-squared error of the predicted spectra to quantify the timescale of predictions and model accuracy respectively.

In addition, we seek to demonstrate that random forest models can improve UV-Vis spectra prediction in general, to a greater accuracy than neural networks, when the available training data is restricted to a small size, as is the case for the number of publicly available available experimental UV-Vis spectra.

In addition, we aim for the MLP head to improve the accuracy of predicted UV-Vis spectra. We can approximately quantify the reduction in error that is gained, between the raw sTDA 10-vector and the MLP-corrected 10-vector, though these values can only be corrected towards values at a higher level of theory (qsGW-BSE in this case), rather than correction to exact values. As such, there is an accumulation of errors from (i) the baseline error of the high-level theory calculation and (ii) the residual error between the MLP-predicted values and the

qsGW-BSE values. Nonetheless we can quantify the total benefit of the MLP by comparing the RMSE of predicted spectra with and without the use of the MLP correction.

3. LITERATURE REVIEW

Having motivated machine-learned predictions of UV-Vis spectra, as a solution to avoid the intractable scaling laws associated with their ab-initio physics, models of this kind tend to come in two categories: (i) Direct prediction of UV-Vis spectra from given training inputs and (ii) Delta-ML models that learn the adjacent physics quantities associated with the spectra.

In the first case, Urbina et al [2] trained a recurrent, attention-based neural network to predict full UV-Vis spectra directly from SMILE strings. This was trained on 3,172 molecules between "Dataset I" and "Dataset II" (see Subsection 4.2. Data), with median predicted spectrum RMSEs of 0.064 and 0.15 achieved respectively, on each data set. This is definitely the most comparable model to what we are trying to achieve in this paper, which is what motivated our selection of the exact same data set, in order to give a more like-for-like comparison in modeling quality.

In addition, McNaughton et al [3] trained a model on 900 molecules, to predict experimental UV-Vis spectra from TD-DFT (ab-initio) UV-Vis spectra, with a median predicted spectrum RMSE of 0.080 achieved. This too is comparable to our model, though performs a highly computationally expensive step to obtain ab-initio spectra, making it inappropriate for high throughput purposes.

Lastly, there are a variety of models that act to correct excited state energies and oscillator strengths, with particular relevance to this report in Verma et al [4], since they were able to correct sTDA excitation energies to within $\sim$0.2eV MAE to CC2 reference energies. There are further models like Ramakrishnan et al's [5] that corrects TD-DFT excited energies and oscillator strengths towards CC2 reference data, with MAE$\sim$0.1eV (though oscillator strength corrections were notably poorer).

The model in this paper improves on its most comparable model: UV-adVISor [2], by incorporating both molecular structure and sTDA-level excited energies and oscillator strengths into the training data, rather than structure alone. It also employs the Δ-ML concept from Ramakrishnan [5] to correct these cheap excitation energies. Therefore, this model's complexity straddles the space between the physics-unaware UV-adVISor [2] and McNaughton's ab-initio model [3]. Despite this, spectrum prediction runtimes of tens of seconds make this paper's model viable for high-throughput screening of UV-Vis spectra.

4. METHODOLOGY AND DATA

4.1. Model Diagram

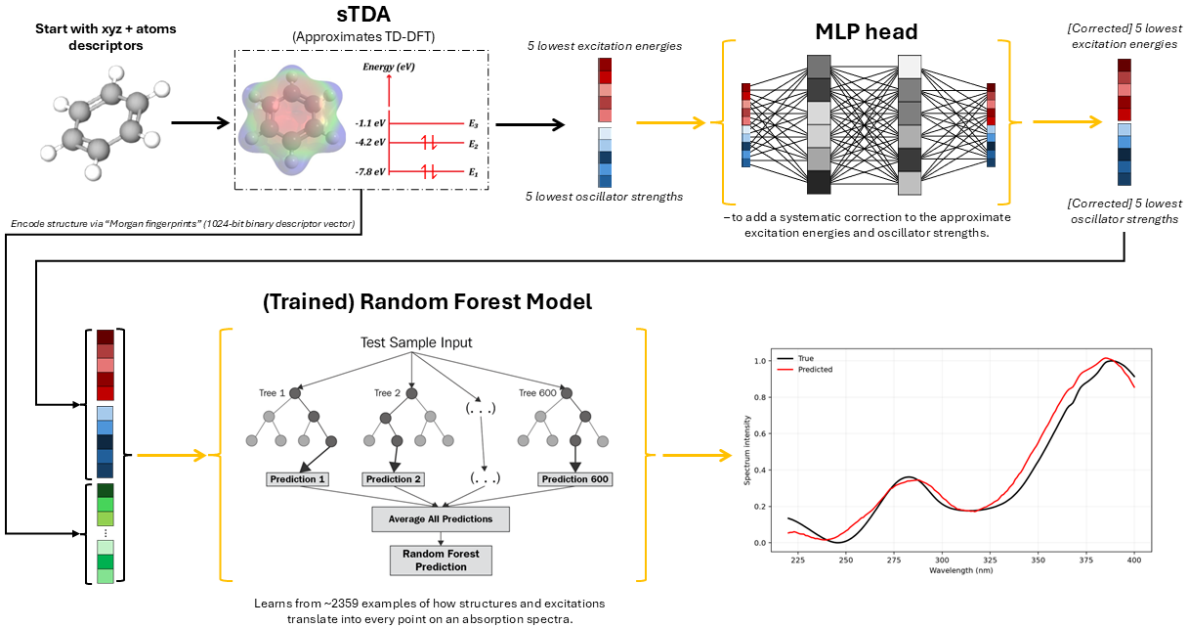


Figure 2. Flowchart of how this paper's machine-learned model predicts UV-Vis spectra in seconds, starting from molecular structure alone.

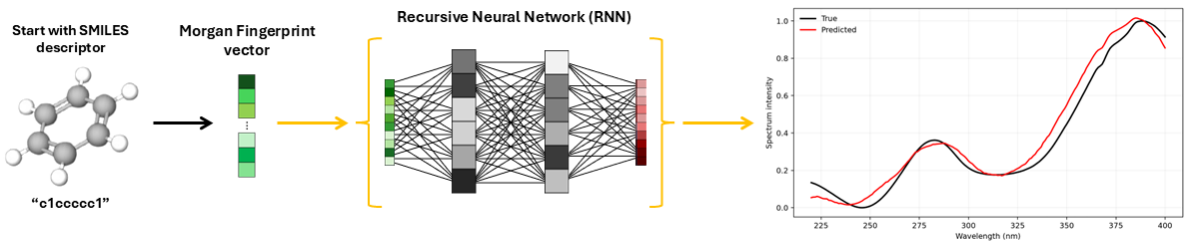


Figure 3. Simple flowchart of the UV-adVISor model [2] for spectrum prediction.

4.2. Data

The two data sources used to train this model are: (i) QM9 [10] - Used to train the MLP head: for the Δ-ML correction to approximate sTDA excitation energies. (ii) UV-adVISor's Dataset I and II [2] - used to train the random forest model for predicting absorption spectra.

N.B. (ii) Was intentionally selected to show the systematic improvement in spectra prediction that is gained by incorporating some quantum calculation, and thereby to draw a direct comparison with the results of UV-adVISor, which just uses SMILE strings as input.

4.3. MLP Head

The MLP (Multi-Layer Perceptron) is a neural network that learns to correct the 10-vector (of 5 excitation energies and 5 oscillator strengths) that is produced by sTDA, towards a reference standard. This Δ-ML concept is used in [5] to correct semi-empirical excitations to approximately DFT-accuracy. It is trained with reference to the QM9 dataset [10] of qsGW-BSE excited energies and oscillator strengths, which is understood to be of a comparable accuracy to hybrid TD-DFT [11]. Its dimensionality is: 10 $\rightarrow$ 512 $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 10 where hidden layers use ReLU activation, and training uses the Adam optimiser [12]. Having been trained, the RMSEs of excitation energies and oscillator strengths were reduced to 0.276eV and 0.066 respectively, with reference to the QM9 dataset.

4.4. sTDA

sTDA (simplified Tamm-Dancoff Approximation) [9] is an efficient excited states method, based on TD-DFT, with the added assumptions that only single excitations are considered in the linear response equations, and atoms interact as monopoles. The original work [9] reports $\sim 2-3x\times$ orders of magnitude greater efficiency over full TD-DFT. This makes it highly suitable to run high-throughput excitation calculations on a large dataset of organic molecules, some up to 126 atoms in size. It is designed to output the 5 lowest-energy electronic transition energies of a given molecule, and their corresponding oscillator strengths; thus outputting a 10-value vector object.

4.5. Encoding SMILES

To efficiently convert a SMILE string into a numerical descriptor of a given molecule's structure, we use "Morgan fingerprints": a vector containing binary values to classify whether the molecule has certain structural traits or not. The authors of the UV-adVISor paper [2] found that 1024-bit encoding performed best in their attention-based model for spectra prediction. Therefore, we have followed suit whenever using SMILE strings as part of training data. It should be noted that combining the 10-dim quantum information vector from sTDA, with a 1024-dim structure vector, can lead to dominant feature-splitting on the much larger vector, potentially failing to sufficiently sample the smaller vector.

4.6. Configuring the Random Forest Model

Random forest models work by sorting molecules down decision trees, dividing them based on their features. A forest is when we generate many decision trees that all divide up molecules via different criteria, such that trees that sort most effectively can be kept and others dispatched with. The termini of the decision trees are called leafs, each of which will identify with one or more training spectra. When we pass down a test molecule down all of the decision trees, it will identify with a different leaf on each tree, at which point we average out the spectra of these leaves' training molecules to get our final result. A variety of Random Forest hyperparameters were experimented with to obtain the best model (see Figures 6 and 7). These hyperparameters correspond to the number of trees in a forest (`n_estimators`), decisions per tree (`max_depth`), how many datapoints for each molecule are inspected at each decision split (`max_features`) and others. Whilst the deep technicalities of Random Forest configuration are outside of this investigation's scope, it is interesting to see in Figures 6 and 7 how each configuration has strengths with certain input data types.

5. RESULTS

The Random Forest model that achieved the lowest median RMSE across both sets is the "deep unregularised" configuration. As such, Figure 4 shows four different sets of inputs that can train the model and their results:

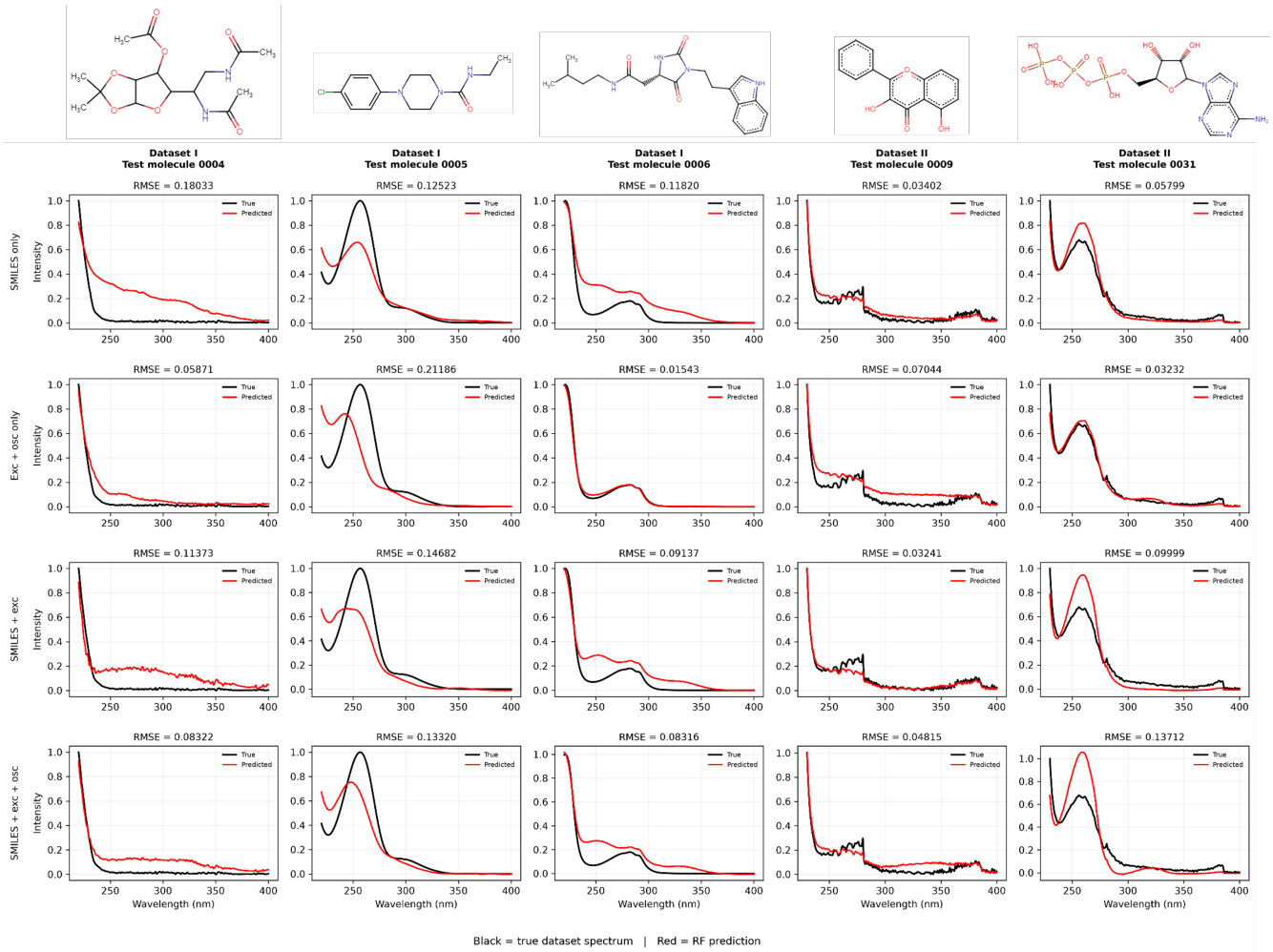


Figure 4. Comparison of experimental (Dataset I and II) UV-Vis spectra alongside their predictions, where these molecules were selected to show a reasonably diverse range of spectrum shapes.

Input Data	RMSE (Dataset I)	RMSE (Dataset II)
SMILES only	0.1198	0.0715
Excitation Energies + Oscillator Strengths	0.1115	0.0784
SMILES + Excitation Energies	0.1066	**0.0791**
SMILES + Excitation Energies + Oscillator Strengths	0.1087	0.0760

Table 1. Median root mean squared error of predicted spectra from the respectively strongest random forest model, given different training inputs. Bold values signify models that employed the MLP. This quantifies the overall performance of the models shown in the rows of Fig. 4.

Model from Paper	Input Data	$RMSE_I$	$RMSE_{II}$	Run time / mol
UV-adVISor [2]	SMILES only	0.064	0.170	< 1 s
This Paper	SMILES, excitation energies, oscillator strengths	0.076	0.109	∼1-100 s
McNaughton [3]	QM spectra, atomic coordinates	∼0.080		∼100-1000 s [7]

Table 2. Median root mean squared error and runtime of predicted spectra from the best models in different literature papers. N.B. runtime refers to the time taken to produce the relevant inputs for each model, and for [3] is inferred from the benchmarked wall time for constructing a UV-Vis spectrum using TD-DFT, which [3] used.

Method	RMSE(Excited Energies) [eV]	RMSE(Oscillator Strengths)
Raw sTDA	0.683	0.087
sTDA + MLP Correction	0.276	0.066

Table 3. RMSE comparison of raw sTDA and MLP-corrected excited-state predictions.

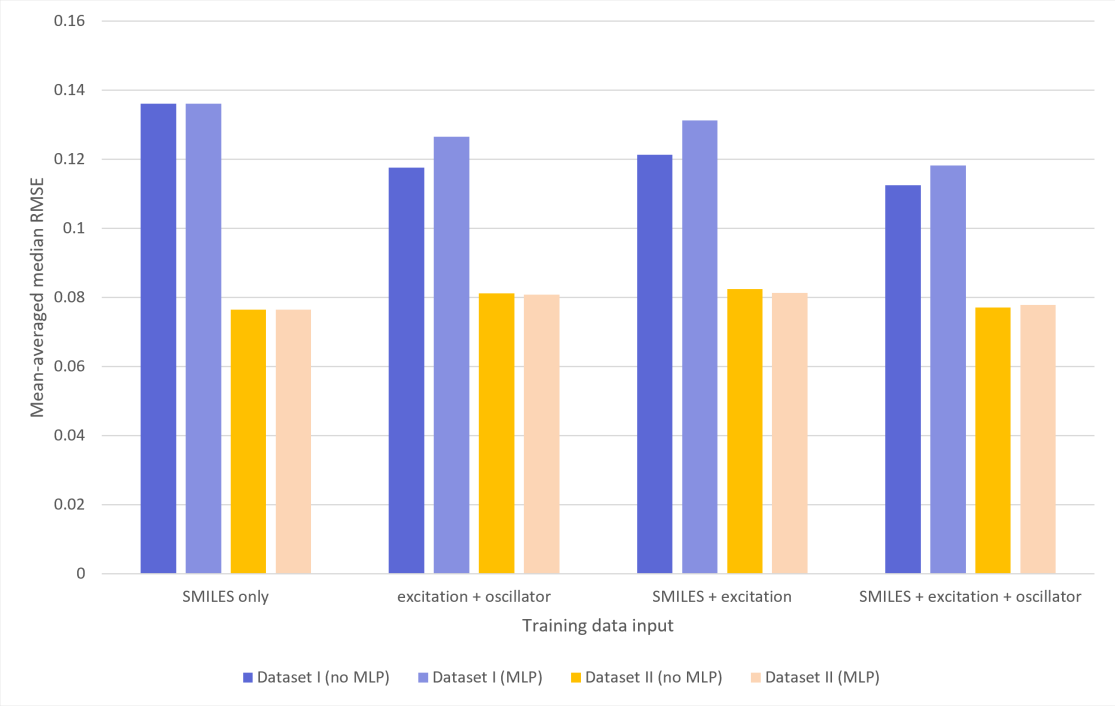


Figure 5. Mean-averaged test set median RMSE values across various random forest models, depending on training data input and, crucially, whether the MLP head was used in training.

Configuration	SMILES only	Excitation + Oscillator	SMILES + Excitation	SMILES + Excitation + Oscillator
No MLP	0.136114	0.117629	0.121356	0.112435
MLP	0.136114	0.126524	0.131309	0.118232

Table 4. RMSE values for Dataset I, mean-averaged across all random forest models, comparing performance with and without the MLP head.

Configuration	SMILES only	Excitation + Oscillator	SMILES + Excitation	SMILES + Excitation + Oscillator
No MLP	0.076413	0.081212	0.082459	0.077105
MLP	0.076413	0.080863	0.081263	0.077782

Table 5. RMSE values for Dataset II, mean-averaged across all random forest models, comparing performance with and without the MLP head.

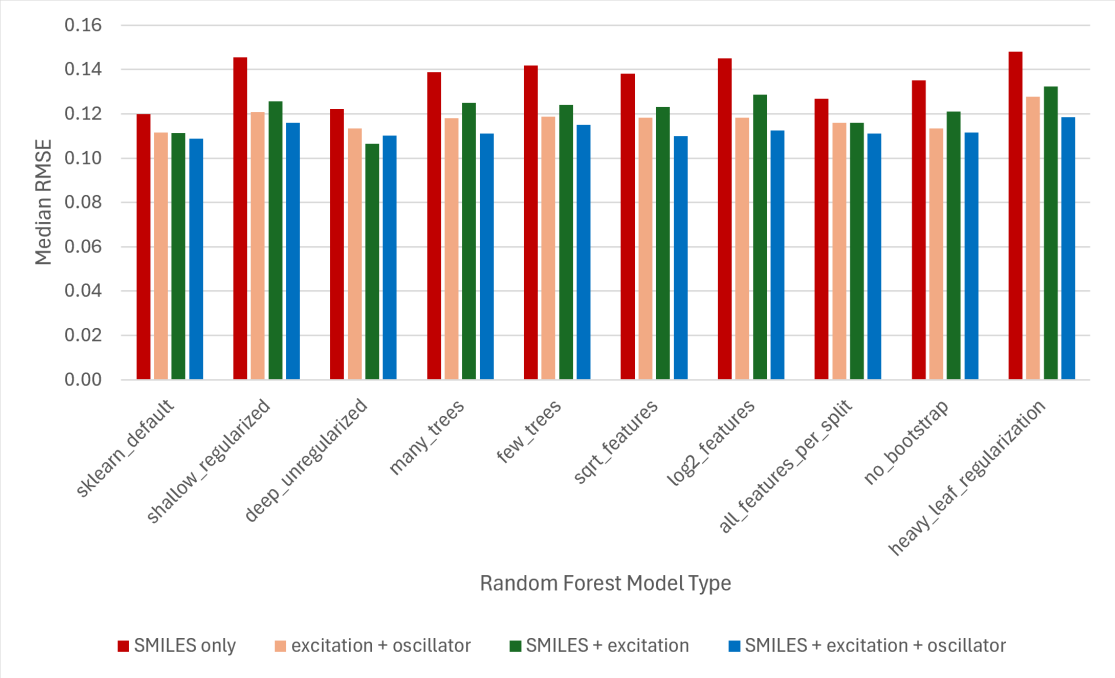


Figure 6. RMSE comparison for Dataset I across different random forest configurations and input data types.

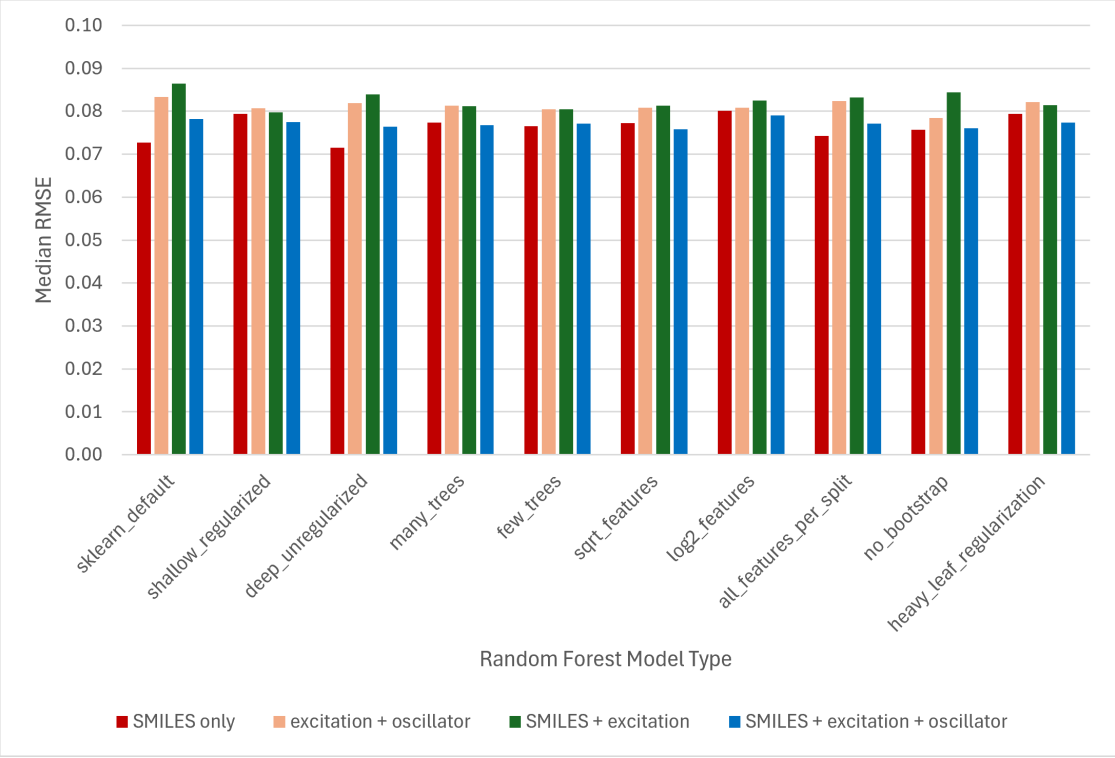


Figure 7. RMSE comparison for Dataset II training across different random forest configurations and input data types.

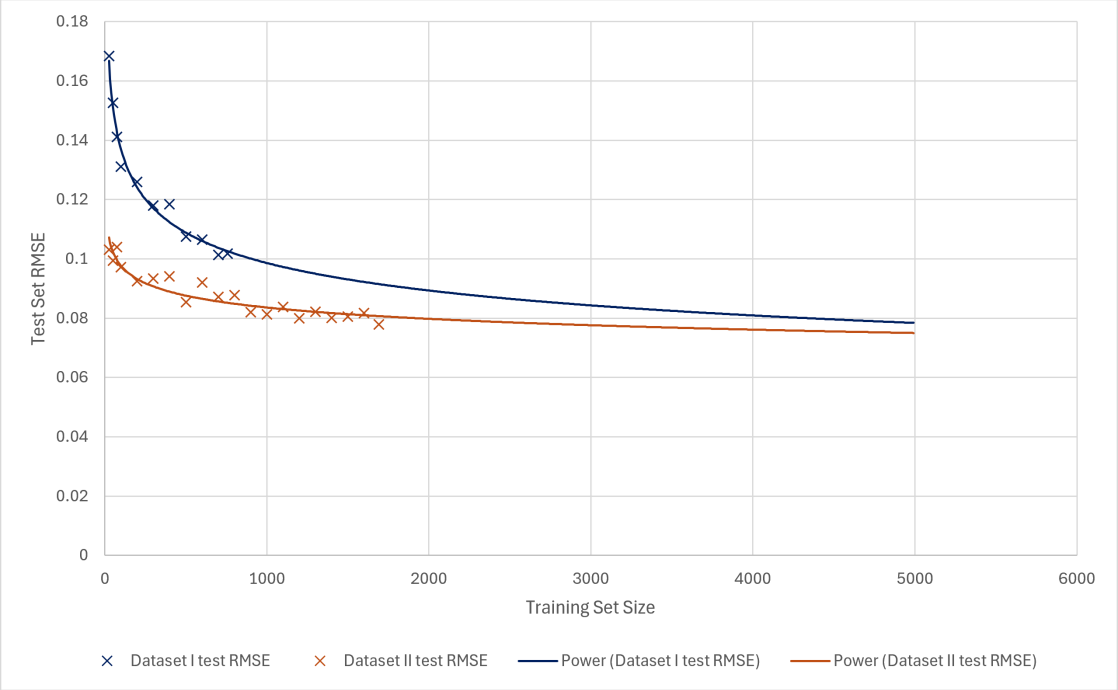


Figure 8. RMSE projected as an inverse power function of training set size, as is standard practice [8].

6. DISCUSSION

This model's overall relative performance is quantified summatively in Table 2. We observe that the combined use of excitation energies, oscillator strengths and a random forest configuration allow this model to obtain lower RMSE in UV-Vis spectra prediction compared to [2] (albeit with longer calculation time), and comparable RMSE to [3] but in lesser calculation time. A caveat is that [3] was trained on a different dataset, but nonetheless provides an interesting benchmark of RMSE = 0.080 for a machine-learned UV-Vis spectrum predictor that uses high-level ab-initio spectra as training input.

An important observation is captured in Figures 6 and 7. That is, that the addition of the cheap sTDA excited states calculation generally lowers the RMSE of UV-Vis spectra predictions across both datasets, by comparison to SMILES-only training. This is more pronounced in Dataset I, which the authors of [2] note is more chemically and structurally diverse. The model's performance on Dataset II is then comparable in accuracy to both [2] and general RMSE achieved on a larger dataset in [3]. This indicates that excited states and oscillator strengths have non-trivial, additional training information and thus predictive power, beyond SMILE strings alone. This makes sense, since the relationship between molecular structure and these physical quantities is extremely complex, and so machine learning is unlikely to capture excitation energies and oscillator strengths accurately, in the absence of exceedingly large model complexity and training set size. We also know (as per our introductory notes) that excitation energies and oscillator strengths can be used to construct UV-Vis spectra directly.

Furthermore, we observe in Figure 4 that the models trained on excitation energies and oscillator strengths alone, performed exceptionally well compared to the other models at predicting the spectra of Dataset I Test molecule 0004 and 0006, and Dataset II test molecule 0031 (which don't have common names except for 0031, which is ATP). These may be examples of molecules where vibrational and solvent effects are relatively less pronounced. Given that both SMILES-only and (excitation energies + oscillator strengths)-only models have some mutually-exclusive sources of error, the combination of them in the SMILES + excitation energies + oscillator strengths must partially cancel these errors and thus produces the most accurate result. All of that said, there also must be some extent to which a machine learning model can "infer" excited state physics, meaning that the combination of SMILES with excitations and oscillator strengths in training data is partially redundant. There is a somewhat open question as to what minimum RMSE could is achievable with maximal quantum

training information, though McNaughton [3] provides an interesting heuristic of RMSE ≥ 0.080 for various neural network models, trained on $\sim$ 1000 DFT-calculated UV-Vis spectra. We note that DFT misses certain physics information though - like vibronic structure, conformational-averaging, approximate solvent interactions etc.

Given that Dataset II is less diverse, and that the authors of [2] used a convolutional neural network achieve a superior error of ~ 0.064 compared to this paper's 0.0715, we may attribute this to the susceptibility of neural networks to memorise noise in small datasets, with random forests being less susceptible to overfitting. As such, if Dataset II spectra are more mutually similar, a neural network will be able to drive error down via a stronger ability to memorise, despite a lesser ability to generalise. This would also explain this neural network's significantly greater error in Dataset I, which is more structurally diverse and therefore puts a greater demand on model generalisation.

In assessment of the machine learning architecture, the choice to use a random forest as the acting model for UV-Vis spectra prediction is vindicated in Table 1, given that it was able to achieve close to the same RMSE (0.0715) as [2] in Dataset II, and superior RMSE (0.1066) in Dataset I, where both were trained on the same number of SMILE strings in the training set. This is somewhat expected, since Random Forests often perform advantageously over neural networks for smaller sets of training data [14, 15]. That said, it is evident from Figure 5 and Tables 4 and 5 that the MLP head provided little-to-no marginal reduction in the RMSE of predicted spectra. This is an interesting observation, considering that in Table 3 we see that the MLP corrections better-approximated qsGW-BSE energies and oscillator strengths to a significant degree. The fact that this doesn't translate to more accurate predicted spectra could partially be down to an aggregation of error from the qsGW-BSE dataset itself, followed by the MLP's error when fitting sTDA values to these. More likely though, is that since sTDA errors have a systematic component, the random forest may essentially be able to adjust these itself. Or rather, it will learn to predict spectrum features by applying its own biases to the input data. In addition, adjusting sTDA values towards qsGW-BSE values inherently involves adding noise to the predictions, which inhibits the random forest's ability to learn these systematic errors in the input data.

Moreover, the limiting factor in the accuracy of this model (as [2] also notes) seems to be the lack of publicly available experimentally-measured UV-Vis spectra. 3,063 data points in the training set is clearly small for machine-learning a function as complex as a UV-Vis spectrum. This limitation is illustrated in Figure 8, which projects how the test set error of a random forest model varies with training set size. It's thus projected that training this model on a data set of $\sim$ 20,000 molecules could approximate RMSE of 0.060 to a general set of molecules. One source of extension in this regard would be using Wiley's commercial dataset of $\sim$ 30,000 UV-Vis spectra [16]. There are also further available sources like NIST ($\sim$1,600 spectra) [17], MPI-Mainz ($\sim$1,402 spectra [18]).

Importantly, we note that we did not include 3/950 of the Dataset I molecules, nor 106/2222 of the Dataset II molecules as part of the training process. This was either due (i) Errors associated with encoding their structure into Morgan Fingerprints, (ii) Errors with accounting for charges or (iii) Simulation time-out, which was set at 3 hours / molecule. This may incrementally harm the accuracy of both sets, since fewer training examples will inhibit model generalisation. Or indeed, it could even artificially improve this model's results, since only the largest of molecules in Dataset II were timed-out (and therefore excluded from the analysis and predictions), which we would expect to have greater associated errors.

Lastly, it is reasonable to conclude that this model offers a fast and (relative to the literature) accurate method for predicting absorption spectra and UV-Vis spectra generally. It is able to generate predictions on the timescale of seconds and does so with an intermediate accuracy between SMILES-only input models and models trained on ab-initio spectra, with particular gains over SMILES-only training on the more diverse Dataset I. With training on a wider range of the UV-Vis wavelength band, vast numbers of candidate OLED molecules could be fitness-tested using this method.

7. DECLARATION

Use of AI: [AI was used in this assessment to configure sTDA calculations, parallelise tasks, generate select simple functions and debug.]

References

[1] X. Yang, H. Ma, Q. Lu, and W. Bian, "Efficient Method for Numerical Calculations of Molecular Vibrational Frequencies by Exploiting Sparseness of Hessian Matrix," *The Journal of Physical Chemistry A*, **128**, 3024-3032 (2024). DOI: 10.1021/acs.jpca.3c07645.

[2] F. Urbina, K. Batra, K. J. Luebke, J. D. White, D. Matsiev, L. L. Olson, J. P. Malerich, M. A. Z. Hupcey, P. B. Madrid, and S. Ekins, "UV-adVISor: Attention-Based Recurrent Neural Networks to Predict UV-Vis Spectra," *Analytical Chemistry*, **93**, 16076-16085 (2021). DOI: 10.1021/acs.analchem.1c03741.

[3] A. D. McNaughton, R. P. Joshi, C. R. Knutson, A. Fnu, K. J. Luebke, J. P. Malerich, P. B. Madrid, and N. Kumar, "Machine Learning Models for Predicting Molecular UV-Vis Spectra with Quantum Mechanical Properties," *Journal of Chemical Information and Modeling*, **63**, 1462-1471 (2023). DOI: 10.1021/acs.jcim.2c01662.

[4] S. Verma, M. Rivera, D. O. Scanlon, and A. Walsh, "Machine learned calibrations to high-throughput molecular excited state calculations," *The Journal of Chemical Physics*, **156**, 134116 (2022). DOI: 10.1063/5.0084535

[5] R. Ramakrishnan, M. Hartmann, E. Tapavicza, and O. A. von Lilienfeld, "Electronic spectra from TDDFT and machine learning in chemical space," *The Journal of Chemical Physics*, **143**, 084111 (2015). DOI: 10.1063/1.4928757.

[6] K. Ghosh, A. Stuke, M. Todorović, P. B. Jørgensen, M. N. Schmidt, A. Vehtari, and P. Rinke, "Deep Learning Spectroscopy: Neural Networks for Molecular Excitation Spectra," *Advanced Science*, **6**, 1801367 (2019). DOI: 10.1002/advs.201801367.

[7] M. Martyka, X.-Y. Tong, J. Jankowska, and P. O. Dral, "OMNI-P2x Universal Neural Network Potential for Excited-State Simulations," *Nature Communications*, **17**, 4949 (2026). DOI: 10.1038/s41467-026-71380-5.

[8] L. J. Frey and D. H. Fisher, "Modeling Decision Tree Performance with the Power Law," *Proceedings of the Seventh International Workshop on Artificial Intelligence and Statistics*, **R2** (1999).

[9] S. Grimme, "A Simplified Tamm-Dancoff Density Functional Approach for the Electronic Excitation Spectra of Very Large Molecules," *The Journal of Chemical Physics*, **138**, 244104 (2013). DOI: 10.1063/1.4811331.

[10] D. Baum, A. Förster, and L. Visscher, "qsGW Quasiparticle and GW-BSE Excitation Energies of 133,885 Molecules," *Scientific Data*, **13**, 643 (2026). DOI: 10.1038/s41597-026-07018-4.

[11] X. Gui, C. Holzer, and W. Klopper, "Accuracy Assessment of GW Starting Points for Calculating Molecular Excitation Energies Using the Bethe-Salpeter Formalism," *Journal of Chemical Theory and Computation*, **14**, 2127-2136 (2018). DOI: 10.1021/acs.jctc.8b00014.

[12] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," *3rd International Conference on Learning Representations (ICLR)*, 2015.

[13] P. J. Linstrom, "NIST Chemistry WebBook, NIST Standard Reference Database Number 69," *National Institute of Standards and Technology*, Gaithersburg, MD. DOI: 10.18434/T4D303

[14] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on tabular data?", in *Advances in Neural Information Processing Systems*, vol. 35, 2022.

[15] S. Silvey and J. Liu, "Sample Size Requirements for Popular Classification Algorithms in Tabular Clinical Data: Empirical Study," *Journal of Medical Internet Research*, **26**, e60231 (2024). DOI: 10.2196/60231.

[16] John Wiley & Sons, Inc., "KnowItAll UV-Vis Spectral Library Collection," *Wiley Science Solutions*. Available: https://sciencesolutions.wiley.com/solutions/technique/uv-vis/ (accessed August 2026).

[17] V. Talrose, A. N. Yermakov, A. N. Leskin, A. A. Usov, A. A. Goncharova, N. A. Messineva, N. V. Usova, M. V. Efimkina, and E. V. Aristova, "UV/Vis Database," in *NIST Chemistry WebBook, NIST Standard Reference Database Number 69*, National Institute of Standards and Technology, Gaithersburg, MD. DOI: 10.18434/T4D303.

[18] H. Keller-Rudek, G. K. Moortgat, R. Sander, and R. Sörensen, "The MPI-Mainz UV/VIS Spectral Atlas of Gaseous Molecules of Atmospheric Interest," *Earth System Science Data*, **5**, 365-373 (2013). DOI: 10.5194/essd-5-365-2013.